

Mingchen Song¹, Xiang Deng¹, Zhiling Zhou¹, Jie Wei¹, Weili Guan¹, and Liqiang Nie¹

¹Affiliation not available

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A Survey on Diffusion Policy for Robotic Manipulation: Taxonomy, Analysis, and Future Directions

Mingchen Song, Xiang Deng*, Zhiling Zhou, Jie Wei, Weili Guan*, Liqiang Nie

Abstract—Robotic manipulation tasks face significant challenges in learning effective control policies in complex, high-dimensional action spaces and uncertain environments. Recently, diffusion models—originally developed for image generation—have emerged as a promising approach for robotic policy learning due to their exceptional ability to model complex multimodal distributions and generate diverse, high quality outputs. Since 2022, numerous research studies have explored diffusion policies for robotic manipulation, demonstrating superior performance compared to traditional methods in various tasks. However, there is a lack of comprehensive survey papers that thoroughly analyze this rapidly evolving field. To bridge this gap, we provide a comprehensive review of diffusion policies for robotic manipulation in a taxonomy of three key dimensions: **1) robot data representations**, including 2D representations (robot trajectories and human videos), 3D representations (single-view and multi-view point clouds), and heterogeneous data (input data and robot proprioception heterogeneity); **2) model architectures**, examining various combinations such as Large Language Models with diffusion head (general data pre-training and robot data pre-training), small-size CNN or transformer models with diffusion (for direct action generation, subgoal generation, robot data generation, and MoE architectures), and VAE/VQ-VAE integrated with diffusion; **3) diffusion strategies**, covering approaches that incorporate reinforcement learning, equivariance, accelerated sampling or denoising strategies, employing classifier (free) guidance, and integration with self-supervised learning. Furthermore, we provide a comprehensive analysis of their technical characteristics, advantages, and limitations. Based on our comprehensive organization and analysis, we offer valuable suggestions for selecting appropriate diffusion policy methods for different robotic manipulation scenarios and outline promising future research directions. To facilitate further research in this field, we maintain a publicly accessible repository at <https://github.com/HITSZ-Robotics/DiffusionPolicy-Robotics>, which serves as an exhaustive resource of diffusion policies for robotic manipulation papers and corresponding open-source codes, and we have meticulously collected and organized the configurations of simulation platforms and real-world robots. We will keep updating this repository to include the latest advancements in the field.

Index Terms—Diffusion policy, Robotic manipulation, Vision-language action generation, Diffusion architectures.

1 INTRODUCTION

Over the past several decades, robotic manipulation has evolved substantially, enabling robots to perform increasingly complex tasks in various environments. Manipulation represents a fundamental capability for robots operating in object-centric environments [1], [2], [3], [4] through the process of interacting with objects to achieve goal-oriented tasks. Traditional approaches including model-based control [5], [6], [7], imitation learning [8], [9], [10], and reinforcement learning [11], [12], [13], [14], have made significant progress in this domain. While these methods have achieved success in constrained scenarios, they collectively face limitations in generalization, sample efficiency, and handling the inherent multimodality of manipulation tasks. Consequently, they often prove impractical for real-world deployment. The development of robust and generalizable policies remains challenging due to several factors: high-dimensional action spaces, complex scene understanding requirements, and uncertain object properties. These challenges are particularly prominent in settings where robots

must handle novel objects and adapt to environmental changes.

Recent years have witnessed remarkable advances in generative modeling through diffusion models [15], [16], [17], [18], [19], which learn to transform simple noise distributions into complex data distributions via iterative denoising. Originally developed for image generation, diffusion models have demonstrated extraordinary capabilities across various domains, making them particularly appealing for robotic manipulation where multiple valid action sequences may exist for a given task. Unlike GANs [20], [21], [22] or VAEs [23], [24], [25], diffusion models offer more stable training processes and superior quality in modeling complex distributions. They operate by gradually denoising samples drawn from a simple distribution (typically Gaussian): the forward process progressively adds noise according to a predefined schedule, while the reverse process learns to denoise corrupted samples step by step. When adapted to robotic manipulation, this enables the generation of action sequences that satisfy task constraints while maintaining diversity, providing a principled approach to handling uncertainty and multimodality in action spaces. Since 2022, as illustrated in Fig.1 and Fig.2, research applying diffusion models to robotic manipulation has rapidly grown, with papers addressing various aspects of this emerging paradigm, collectively called “diffusion policy.”

• *Corresponding author.
Mingchen Song, Xiang Deng, Zhiling Zhou, Jie Wei, Weili Guan and Liqiang Nie are with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), China, 518055 (e-mail: {mingchens0905, xdeng2023, zzl3230136428, wjl83163, honeyguan, nieliqiang}@gmail.com).

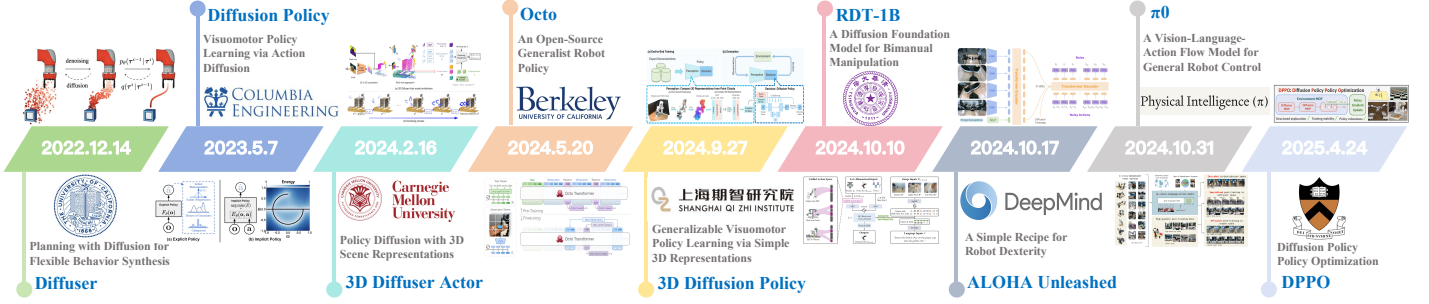


Fig. 1: Timeline of notable developments (non-exhaustive) in diffusion policies for robotic manipulation.

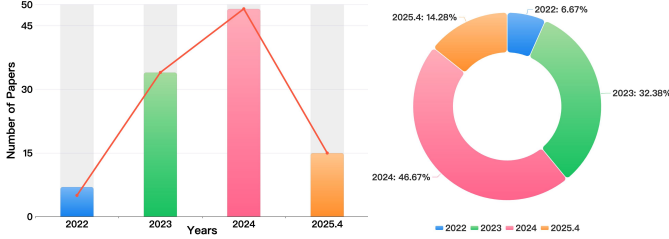


Fig. 2: The number of papers on diffusion policies for robotic manipulation changes over time, with a deadline of April 2025.

The rising interest in diffusion policies for robotic manipulation can be attributed to several key advantages over conventional approaches. First, in terms of multi-modal distribution modeling, diffusion models are excellent at capturing multi-modal action distributions [26], [27], [28], [29], [30]. This enables robots to represent multiple valid strategies for the same task. For example, in robotic manipulation scenarios where an object can be grasped from different angles or alternative paths can be followed to avoid obstacles, this capability is highly practical and valuable. Second, with respect to sample efficiency, diffusion policies can reduce data requirements compared to traditional reinforcement learning approaches by learning from demonstrations or limited interactions [31], [32], [33]. The generative modeling framework allows for effective learning from offline datasets, making diffusion policies especially suitable for scenarios where real-world robot trajectory data are costly or limited. Third, in relation to generalization capabilities, the flexible representation capabilities of diffusion models allow for better generalization across task variations [28], [34], [35], object appearances [31], and environmental conditions [36], [37], [38], [39]. By modeling and learning manipulation behaviors instead of memorizing specific trajectories, diffusion policies can adapt to novel scenarios more effectively. Finally, for long-horizon planning, recent advancements have demonstrated that diffusion policies can generate coherent long-horizon action sequences [32], [40], [41], [42], which addresses a significant challenge in robotic manipulation. The iterative denoising process enables considering past states, generating temporally consistent actions while maintaining physical plausibility throughout extended manipulation sequences. This temporal consistency proves essential for executing sophisticated multi-step tasks that require maintaining contextual awareness across extended time horizons.

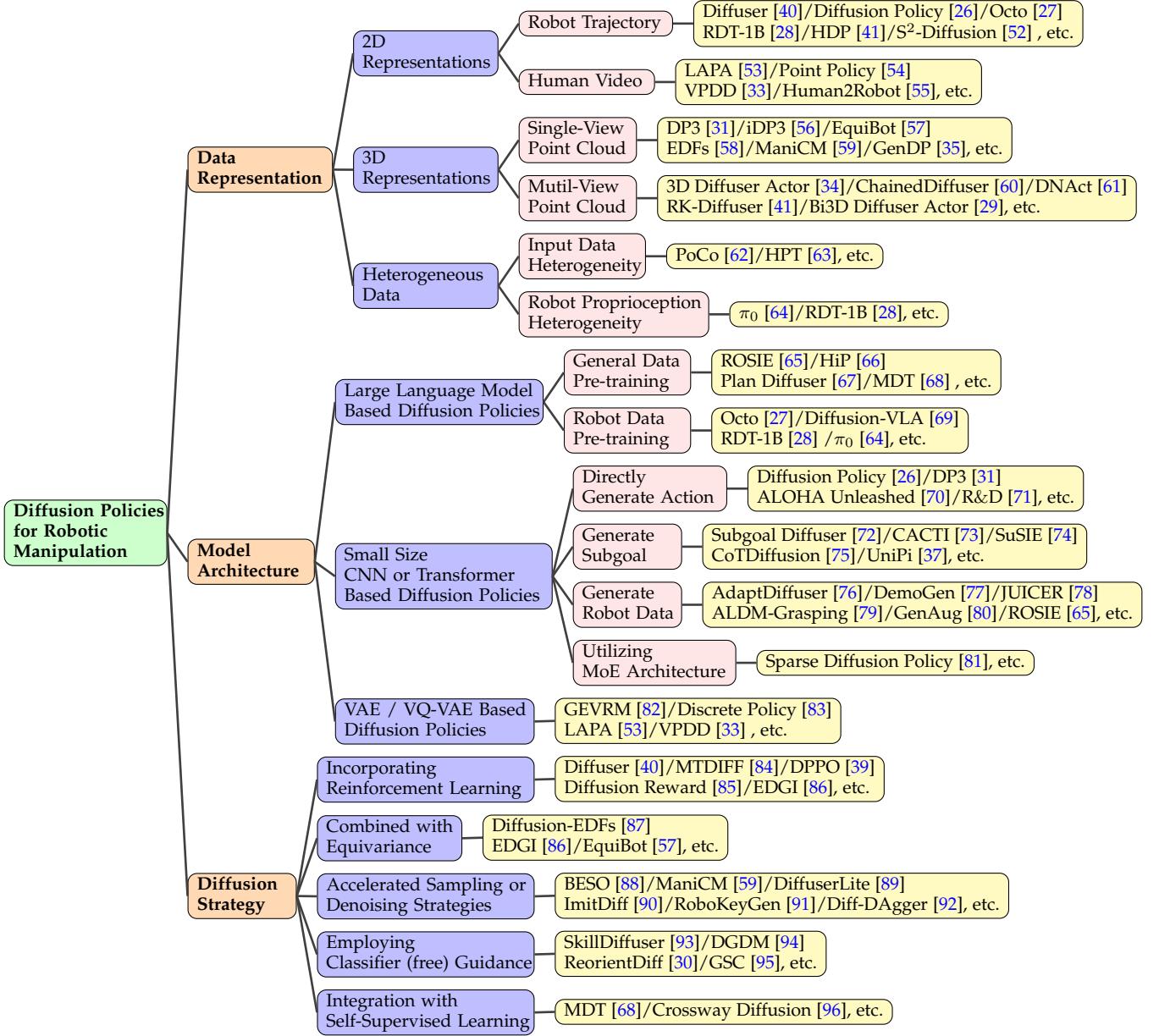
However, despite the rapidly growing body of literature on diffusion policies for robotic manipulation, there is a lack of systematic organization and comprehensive analysis of this emerging field. Existing surveys on robotic manipulation [43], [44], [45], [46] primarily focus on traditional approaches and do not cover recent developments in diffusion policies. Similarly, reviews on diffusion models [47], [48], [49], [50], [51] typically emphasize applications in computer vision and natural language processing, with limited attention to robotics applications. The absence of a dedicated survey makes it challenging for researchers and practitioners to navigate this rapidly evolving field, understand the relationships between different methods, and identify promising research directions.

In summary, we aim to bridge this gap by providing a comprehensive survey of diffusion policies for robotic manipulation. As shown in Table 1, our work comprehensively categorizes existing approaches from different perspectives, analyzes their methodological foundations, advantages, characteristics, limitations, and proposes valuable recommendations and future directions. By systematically organizing and analyzing the related work in this emerging direction, we seek to accelerate progress in developing more capable, robust, and generalizable diffusion policies for robotic manipulation.

Our contributions can be summarized as follows:

- **Systematic taxonomy and comprehensive review:** This paper presents the first systematic categorization and comprehensive analysis of diffusion policies for robotic manipulation. We establish a taxonomy organized along three fundamental dimensions: robot data representations, model architectures, and diffusion strategies. For each category, we conduct a thorough analysis of various diffusion policy methods, highlighting their respective strengths and limitations, offering a holistic overview of the current diffusion policies landscape.
- **Recommendations and Future Directions:** We provide practical and valuable guidance for selecting appropriate methods based on task characteristics, available data, computational constraints, and other practical needs. Furthermore, we identify promising research directions for diffusion policies for robotic manipulation, including theoretical questions about diffusion-based action generation foundations, methodological innovations, and practical considerations for real-world deployment.

TABLE 1: Taxonomy of diffusion policies for Robotic Manipulation



- **Collection of abundant resources:** We maintain a public repository at <https://github.com/HITSZ-Robotics/DiffusionPolicy-Robotics> serving as a comprehensive resource for diffusion policy papers and implementations, including links to papers, open-source code, simulation benchmarks, and real-world robots to facilitate reproducibility and fair comparisons. We will continuously update our website based on the latest developments in diffusion policies for robot manipulation, ensuring that researchers have access to current and relevant resources in this rapidly evolving field.

The remainder of this survey is organized as follows. Section 2 begins with the energy-based models, exploring their applications in robotic manipulation. We then discuss their connection to diffusion models and provide an introduction to them. Section 3 analyzes robot data representation approaches, categorizing them into 2D, 3D, and heterogeneous data. Section 4 examines diffusion policy model architectures across three categories: large language models with diffusion, small-size models with diffusion, and VAE/VQ-VAE integrated with diffusion. Section 5 explores various diffusion strategies including reinforcement learning integration, equivariance properties, accelerated sampling, classifier guidance, and self-supervised learning. Section 6 presents major simulation platforms and real-world robots. Section 7 provides suggestions and future directions, followed by conclusions in Section 8.

Section 2 begins with the energy-based models (EBMs [97], [98]) and their fundamental principles, which serve as the mathematical foundation for many generative

2 BACKGROUND

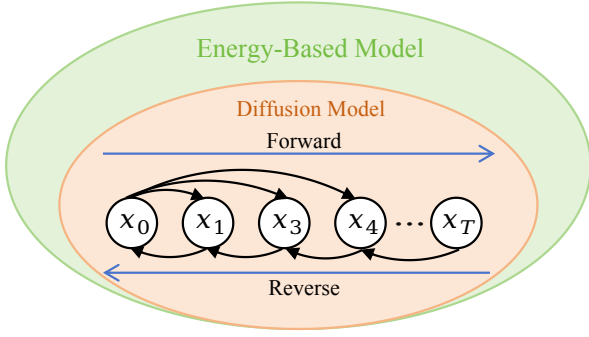


Fig. 3: The relationship between Energy-Based Models and Diffusion Models.

modeling techniques. We then delve into how these energy-based formulations naturally extend to diffusion models through thermodynamic interpretations. Finally, we present a detailed description of diffusion models, including their mathematical principles and improvement methods.

2.1 From Energy-Based Model to Diffusion Model

EBMs provide a powerful framework for modeling complex probability distributions by assigning scalar energy values to variable configurations. These models define probability distributions through energy functions that attribute lower energies to more probable configurations. The probability density function in an EBM is expressed as:

$$p_{\theta}(\mathbf{x}) = \frac{\exp(-E_{\theta}(\mathbf{x}))}{Z_{\theta}}, \quad (1)$$

where $E_{\theta}(\mathbf{x})$ denotes the energy function parametrized by θ , and $Z_{\theta} = \int \exp(-E_{\theta}(\mathbf{x}))d\mathbf{x}$ represents the normalization constant or partition function.

A distinguishing feature of EBMs is their exceptional sample efficiency [58], [99] in learning contexts, often requiring minimal demonstrations (approximately 5-10) to effectively capture underlying data distributions. This characteristic makes them particularly valuable for robotic manipulation tasks where data collection is resource-intensive. For instance, Equivariant Descriptor Fields (EDFs [58]) employ Lie group representation theory [100] to construct SE(3)-equivariant energy-based models that enable end-to-end learning for robotic manipulation from point clouds with remarkably few demonstrations.

The mathematical foundation of EBMs typically involves maximizing the log-likelihood of observed data:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log p_{\theta}(\mathbf{x})], \quad (2)$$

which expands to:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [-E_{\theta}(\mathbf{x})] - \log Z_{\theta}. \quad (3)$$

Computing the gradient of this objective function presents challenges due to the intractable nature of the partition function Z_{θ} . Consequently, alternative training methods are employed, including contrastive divergence [101], Markov chain Monte Carlo [98], or score matching [16], [98]. As Ryu et al. [58] note, EBMs on non-Euclidean manifolds like the SE(3) group [102] require specialized techniques, as

conventional Langevin dynamics [103] cannot be directly applied. Instead, specialized Langevin dynamics formulations for Lie groups are utilized.

Recent research has enhanced EBMs with geometric constraints such as SE(3)-equivariance to improve performance in spatial reasoning tasks. This equivariance property ensures the model respects domain-specific symmetries, leading to superior generalization. Neural Descriptor Fields (NDFs) [104] exemplify this approach by encoding object-target relationships through category-level descriptors, providing representations that facilitate manipulation tasks by encoding points and relative poses.

A fundamental connection between EBMs and diffusion models emerges through thermodynamics and statistical physics principles. As shown in Figure 3, diffusion models can be interpreted as specialized EBMs where the energy function corresponds to free energy during non-equilibrium thermodynamic processes [15], [16], [97]. The diffusion process transforms a simple distribution (typically Gaussian noise) into a complex data distribution by simulating time-reversed diffusion.

From a thermodynamic perspective, the progressive noise addition in diffusion models' forward process increases system entropy [105], while the reverse process decreases entropy to recover the original data distribution. This interpretation aligns with non-equilibrium statistical mechanics principles, where systems evolve according to stochastic differential equations describing particle dynamics under thermal fluctuations.

The score function in diffusion models, defined as the gradient of log probability density with respect to input, establishes a direct connection to EBMs:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\nabla_{\mathbf{x}} E(\mathbf{x}). \quad (4)$$

This relationship demonstrates that estimating the score function in diffusion models is equivalent to estimating the energy function gradient in EBMs, establishing a fundamental connection between these approaches.

2.2 Diffusion Model

Diffusion models [15] stands for a class of generative models that learn to reverse a progressive noise addition process. The framework comprises a forward diffusion process that systematically introduces Gaussian noise to data and a learned reverse process that eliminates this noise. The forward process is formalized as a Markov chain that incrementally adds Gaussian noise according to a predefined variance schedule $\{\beta_t\}_{t=1}^T$:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad (5)$$

Starting from a data point $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, this process generates increasingly noisy samples $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T$, with $\mathbf{x}_T$ approximating Gaussian noise. A key property allows direct sampling of $\mathbf{x}_t$ at any timestep t through reparameterization:

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (6)$$

Here, $\alpha_t = \prod_{i=1}^t (1 - \beta_i)$, which represents the cumulative noise scaling factor.

The reverse process aims to reconstruct the original data distribution by progressively denoising, beginning from

pure noise. This process is also characterized as a Markov chain:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t), \Sigma_{\theta}(\mathbf{x}_t, t)). \quad (7)$$

Under specific conditions, the mean $\mu_{\theta}(\mathbf{x}_t, t)$ can be expressed in terms of a noise prediction network $\epsilon_{\theta}(\mathbf{x}_t, t)$:

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right). \quad (8)$$

The training objective for diffusion models minimizes the variational upper bound on negative log-likelihood, which simplifies to a reweighted mean squared error between predicted and actual noise:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t \sim [1, T], \mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),} [||\epsilon - \epsilon_{\theta}(\sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon, t)||^2]. \quad (9)$$

During inference, sample generation begins by drawing $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and then iteratively applying the learned reverse process to derive $\mathbf{x}_{T-1}, \mathbf{x}_{T-2}, \dots, \mathbf{x}_0$.

Researchers have developed various sampling strategies to enhance efficiency and generation quality, including Denoising Diffusion Implicit Models (DDIM [17]), which enable non-Markovian sampling paths and accelerated generation. Furthermore, classifier guidance techniques [106] can direct the generation process toward specific attributes or conditions.

3 ROBOT DATA REPRESENTATION

The effectiveness of diffusion policies for robotic tasks heavily depends on how robot data is represented and processed. This section examines different approaches to diffusion policies for robotic manipulation from the perspective of robot data representations, categorizing them into three main types: 2D representations (robot trajectories and human videos), 3D representations (single-view and multi-view point clouds), and heterogeneous data (input data and robot proprioception heterogeneity).

3.1 2D Representations

2D representations dominate diffusion policies research in robotic manipulation due to their computational efficiency and relative simplicity. The research in this domain falls into two main categories: methods that directly utilize robot trajectory data and those that leverage human videos.

Robot Trajectory. Using robot trajectory learning for robot motion generation is a simple and direct category of approaches. Diffuser [40] introduces the concept of treating trajectory generation as conditional generative modeling, where behaviors emerge through iterative denoising of state representations. Diffusion Policy [26] extends this concept to visuomotor control by employing temporal CNN or Transformer architectures that process image sequences to denoise action trajectories. Large-scale models such as Octo [27] and RDT-1B [28] unify diverse robot datasets into transformer-based policies with diffusion heads, achieving remarkable cross-task generalization. Architectural innovations follow with DiT-Block Policy [107] identifying essential components for performance stability, and MTDP [108] developing modulated attention modules that better integrate

visual conditions with states. Hierarchical approaches like HDP [41] learn to predict optimal end-effector poses before generating dynamically feasible motion trajectories conditioned on current and predicted poses, while DTP [32] generates 2D trajectories through diffusion models to guide policy learning for long-term tasks. Enhancement techniques emerge with Crossway Diffusion [96] improving visual representations through self-supervised learning, and S²-Diffusion [52] combining semantic understanding with spatial representations for category-level generalization. Specialized applications expand to object manipulation with ReorientDiff [30] generating intermediate reorientation poses, collision avoidance [109] in multi-arm settings, generalized manipulation across varied visual conditions [110], and MPD [38] proposing diffusion models as priors while leveraging inverse denoising processes to sample directly from task goal-conditioned posterior trajectory distributions, accelerating robot motion planning optimization.

Human Video. Human videos offer a promising solution to data scarcity challenges in robotic manipulation with diffusion models. LAPA [53] trains a VQ-VAE based action quantization model [24] to extract discrete latent actions from internet-scale videos without robot labels, then pre-trains a 7B-parameter latent VLA model [111] to predict latent action representations before fine-tuning on limited robot data. Similarly, VPDD [33] compresses both human and robot videos into unified tokens through VQ-VAE, employing discrete diffusion with mask-and-replace strategies to predict future video tokens in latent space, and using imagined future videos to guide low-level action learning during fine-tuning. Taking a different but complementary approach, Human2Robot [55] establishes a novel method for precise alignment between human and robot actions through VR-based data collection, using diffusion models to simultaneously generate robotic videos and corresponding actions from human hand videos. Further advancing this field, Point Policy [54] eliminates the need for teleoperation data entirely by learning directly from offline human demonstration videos, utilizing advanced vision models to transform human gestures into robot poses through semantically meaningful keypoints, creating embodiment-agnostic representations.

Discussion. 2D representation approaches provide a foundation for diffusion policies for robotic manipulation through computational efficiency and convolutional architecture compatibility. Robot trajectory methods leverage precise action labels and structured training data, enabling accurate policy learning but requiring costly data collection. Conversely, human video methods utilize abundant internet demonstrations, reducing costs while exposing models to diverse scenarios, yet face the challenge of bridging the human-robot embodiment gap through domain adaptation or representation learning. This transfer often increases complexity and may reduce performance compared to robot-centric approaches. Both methodologies demonstrate strengths: robot trajectory methods excel in precision while human video approaches offer superior scalability and diversity, suggesting potential benefits from hybrid approaches combining these strengths.

3.2 3D Representations

3D representations enhance diffusion policies with explicit spatial understanding, enabling more effective reasoning about object relationships and manipulation tasks in 3D space. The following subsections analyze single-view and multi-view point cloud methods that advance 3D-aware diffusion policies.

Single-View Point Cloud. Single-view point cloud methods extract 3D geometric information from depth sensors while maintaining computational efficiency. As a groundbreaking contribution in this area, DP3 [31] pioneers this approach with sparse point sampling and an efficient MLP encoder for compact 3D representations, demonstrating superior performance over 2D policies. Building upon this foundation, iDP3 [56] enhances DP3 for general-purpose robots with egocentric 3D visual representations. In parallel, several works incorporate equivariance properties to enhance the model’s generalization capability. Specifically, EquiBot [57] combines SIM(3)-equivariant neural networks with diffusion models to ensure generalization across object transformations, while EDFs [58] propose end-to-end trainable SE(3)-equivariant energy-based models, and furthermore, GenDP [35] introduces 3D semantic fields that combine spatial and semantic information from vision foundation models. Moving toward the specific manipulation scenarios, task-specific approaches emerge with ADAMANIP [112] developing adaptive demonstrations and 3D visual diffusion for articulated object manipulation. In addition, RPDiff [113] addresses 6-DoF relational rearrangement across shapes and scene layouts. Similarly, StructDiffusion [114] combines object-centric transformers with diffusion models for complex 3D planning. In the specialized domain of grasping, DexGrasp-Diffusion [115] presents a unified functional grasp synthesis method for multi-dexterous hands. Correspondingly, DexDiffuser [116] generates high-quality grasps from partial point clouds, and additionally, RoboKeyGen [91] employs score-based diffusion for 3D keypoint generation to estimate robot pose and joint angles. Finally, ManiCM [59] accelerates inference with consistency models that enable one-step action generation directly from point cloud inputs.

Multi-View Point Cloud. Multi-view point cloud approaches integrate information from multiple camera perspectives to construct comprehensive 3D representations, addressing occlusion issues and enabling more precise manipulation. ChainedDiffuser [60] lifts 2D feature maps to 3D using sensed depth. It first generates end-effector poses, and then uses them to connect the current pose to the target pose. Building upon this foundation, RK-Diffuser [41] enhances the framework by incorporating kinematic awareness, predicting next-best end-effector poses at a high level before generating kinematically-feasible action trajectories between poses using diffusion models. Concurrently, DNAct [61] pursues an alternative approach, leveraging NeRF [117] for 3D pretraining to learn unified semantic and geometric representations, utilizing multiple cameras around the robot to capture multi-view images, and employing diffusion training to distinguish features across multi-modal, multi-task demonstrations. Subsequently, 3D Diffuser Actor [34] advances this paradigm by tokenizing 3D scene representa-

tions from multiple posed cameras and sensed depth, generating 3D trajectories conditioned on language instructions and explicitly leveraging tokenized 3D scene representations. More recently, Bi3D Diffuser Actor [29] extends the 3D Diffuser Actor framework to bi-manual manipulation, aggregating 3D scene features from multiple camera views to jointly generate coordinated 3D trajectories for both left and right robot end effectors.

Discussion. The shift from 2D to 3D representations in diffusion policies marks a significant advancement in robotic manipulation by enhancing spatial reasoning through explicit depth encoding and geometric relationship modeling. Single-view point cloud approaches offer practical deployment with minimal hardware requirements and improved performance over 2D policies, though they remain limited by occlusions and restricted perspective. Multi-view methods address these limitations by synthesizing data from multiple cameras, creating comprehensive 3D representations that reduce occlusion problems, albeit with increased system complexity and computational demands. Each approach serves distinct purposes: single-view methods suit resource-constrained settings requiring implementation simplicity, while multi-view approaches address complex tasks demanding thorough spatial understanding. As computational capabilities advance and multi-camera systems become more accessible, these complementary approaches will likely evolve in parallel, serving different needs across robotic manipulation applications.

3.3 Heterogeneous Data

Robot manipulation often involves dealing with heterogeneous data from different sensors, domains, or robot platforms. We categorize heterogeneity in robot manipulation into two types: Input Data Heterogeneity, addressing challenges of integrating diverse sensor inputs and data sources, and Robot Proprioception Heterogeneity, focusing on managing variations in embodiments and control interfaces across different robotic platforms.

Input Data Heterogeneity. Addressing diverse input data sources is a fundamental challenge in robotic manipulation. PoCo [62] introduces a framework for policy composition that integrates information from heterogeneous modalities by combining data distributions represented as diffusion models. This approach enables task-level composition for multi-task manipulation and allows independently trained policies to be combined at test time without retraining, particularly valuable for multi-modal inputs like images, point clouds, and tactile information. Similarly, HPT [63] presents a trunk structure addressing dataset heterogeneity across domains including robot teleop [118], deployed robots, simulations, and human videos [119], while remaining compatible with diffusion policies heads and maintaining consistent performance across modalities.

Robot Proprioception Heterogeneity. The challenge of proprioceptive heterogeneity across robot platforms has inspired several innovative solutions. RDT-1B [28] addresses the heterogeneity problem in bimanual manipulation through a diffusion foundation model. It leverages a transformer backbone and carefully designs a multi-modal encoding strategy to eliminate the heterogeneity

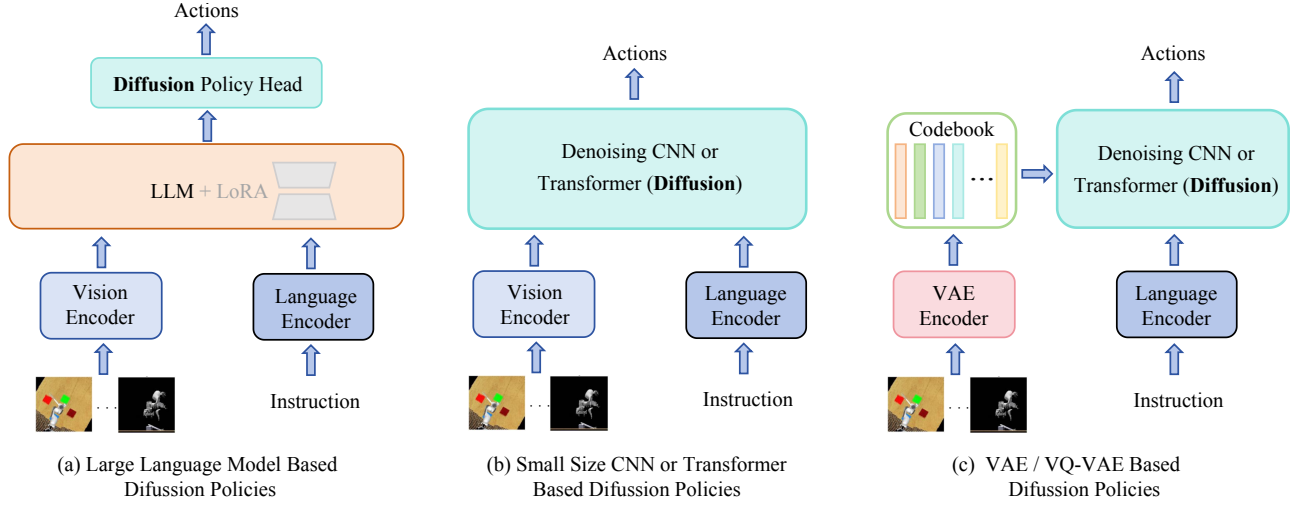


Fig. 4: Comparing architectures of different diffusion policy models. (a) Large Language Model Based Diffusion Policies, (b) Small Size CNN or Transformer Based Diffusion Policies, and (c) VAE / VQ-VAE Based Diffusion Policies.

across different modalities. Moreover, it utilizes an image-text-aligned, pre-trained vision encoder, SigLIP [120], and keeps its weights fixed during training to conserve GPU memory. Similarly, π_0 [64] develops a control model that handles proprioceptive heterogeneity through a pre-trained vision-language model backbone [121] combined with cross-embodiment datasets, using flow matching [122], [123], [124], [125], [126] to generate continuous actions.

Discussion. Addressing heterogeneous data in robotic manipulation represents a critical frontier in developing more adaptable and intelligent robotic systems. The core challenge lies in creating representations and learning strategies that can effectively bridge differences across input data and robot proprioceptions. Successful approaches demonstrate the importance of developing flexible frameworks that can extract universal features while maintaining the ability to specialize for specific robotic platforms.

4 MODEL ARCHITECTURE

In this section, we provide a detailed analysis of different model architectures. As illustrated in Fig. 4, we categorize our analysis into three primary model architectures: (1) Large Language Model Based Diffusion Policies that leverages semantic knowledge from pre-trained models, (2) Small Size CNN or Transformer Based Diffusion Policies that prioritizes computational efficiency, and (3) VAE/VQ-VAE Based Diffusion Policies that focuses on advanced representation learning. Each architectural paradigm presents unique advantages and limitations for manipulation tasks, reflecting the diverse requirements across robotic application scenarios.

4.1 Large Language Model Based Diffusion Policies

The combination between large language models and diffusion techniques represents a pivotal frontier in robotic manipulation, bridging semantic understanding with probabilistic action generation. This integration leverages the rich representational capabilities of pre-trained language models and the inherent generative potential of diffusion

processes. Based on the distinction in pre-training data, we categorize the existing approaches into two primary paradigms: methods utilizing general data pre-training and robot data pre-training.

General Data Pre-training. The general data pre-training category demonstrates pre-training the LLM with general data to leverage the universal knowledge contained in LLMs. MDT [68] leverages pre-trained foundation models such as CLIP [127] and Voltron [128] to master long-horizon manipulation tasks with minimal language annotations. For each task, GPT-4 [129] is queried to generate different text instructions for more diverse language descriptions. Scaling Up and Distilling Down [130] employs LLMs to guide high-level planning and infer success conditions while using diffusion-based policies for multi-task language-conditioned control. Taking a different perspective on data enhancement, ROSIE [65] utilizes state-of-the-art text-to-image diffusion models for aggressive data augmentation, employing large language models to generate text prompts for inpainting unseen objects and backgrounds. TinyVLA [131] freezes the pre-trained multimodal model, utilizing parameter-efficient fine-tuning techniques to enable fast and resource-efficient robot action outputs with only 5% trainable parameters. Moving from data to planning, HiP [66] presents compositional foundation models for hierarchical planning, which represents a comprehensive integration by composing expert models trained on language, vision, and action data, using a large language model for task planning, a video diffusion model for trajectory generation, and a pre-trained inverse model for action mapping. Similar to HiP, Plan Diffuser [67] autoregressively uses LLMs to generate single-step textual sub-goals and employs diffusion models to transform these sub-goals into visual sub-goals for subsequent planning.

Robot Data Pre-training. The robot data pre-training approach enhances robotic manipulation through specialized LLM training on robot-specific datasets, aiming to optimize performance for particular robotic applications. Octo [27] integrates 25 datasets from the Open X-Embodiment Dataset [118], thereby enabling robust generalization across

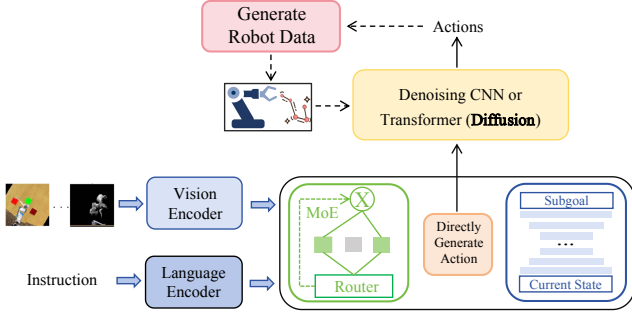


Fig. 5: Small Size CNN or Transformer based Diffusion Policies can be classified into four types: Directly Generate Action, Generate Subgoal, Generate Robot Data and Utilizing MoE Architecture.

diverse robot embodiments and environments. At its foundation lies a transformer architecture with a diffusion head that converts input tokens from observations and tasks into output tokens for action generation. π_0 [64] develops a novel robot policy by leveraging a pre-trained vision-language model backbone, complemented by a flow matching action expert that facilitates precise and smooth manipulation capabilities. Diffusion-VLA [69] adopts a similar methodology, pre-training on the Open X-Embodiment Dataset and Droid dataset [132] and subsequently implementing task-specific fine-tuning using LoRA [133] techniques to optimize the Vision Language Model (VLM). Furthermore, ChatVLA [134] advances the vision-language-action framework by training on robot data while incorporating reasoning datasets and employing staged alignment with MoE architecture to minimize task interference. This approach strengthens visual-textual alignment through co-training, enabling proficiency in both visual-text comprehension and robotic control tasks. In parallel, RDT-1B [28] focuses on specialized manipulation skills, particularly bimanual capabilities. To address the challenge of processing heterogeneous multi-robot data, the approach establishes a unified action space with each dimension corresponding to specific physical attributes. Finally, LAPA [53] further extends this research by pre-training models using both Bridgev2 [135] and the Open X-Embodiment Dataset [118], systematically evaluating cross-embodiment performance and pre-training effectiveness. Collectively, these approaches represent a sophisticated evolution in robotic manipulation, demonstrating the potential of comprehensive data pre-training to expand generalization and advanced manipulation capabilities.

Discussion. The integration of large language models with diffusion techniques represents a significant advancement in robotic manipulation. This approach enables more nuanced and adaptable robotic manipulation by leveraging semantic understanding and probabilistic action generation. The methods demonstrate remarkable capabilities in handling complex, multimodal inputs, ranging from language instructions to visual observations. However, challenges remain in computational efficiency, generalization across different robotic platforms, and maintaining consistent performance across diverse manipulation tasks. Future research may focus on more efficient model architectures and developing more robust transfer learning techniques.

4.2 Small Size CNN or Transformer Based Diffusion Policies

Recent advances in robot manipulation have demonstrated the remarkable potential of combining CNN or Transformer architectures with diffusion models. As shown in Figure 5, based on their technical characteristics and model architecture, we categorize these approaches into four distinct types: methods that directly generate actions, approaches that produce intermediate subgoals, techniques that synthesize robot training data, and architectures that leverage Mixture of Experts (MoE [136]) for enhanced performance.

Directly Generate Action. The first category of methods directly utilizes diffusion models to generate robot actions. Diffusion Policy [26] introduces visuomotor policy learning via action diffusion with both CNN and Transformer architectures, which DP3 [31] extends with point cloud representations for improved spatial generalization. Motion Planning Diffusion [137] applies these concepts to trajectory generation using temporal U-Net architectures, while architectural refinements emerge through DiT-Block Policy [107] identifying key diffusion transformer components and MTDP [108] integrating guiding conditions via modulated attention layers. Similarly, DexDiffuser [116] generates high-quality grasp poses for multi-fingered manipulation by progressively denoising randomly sampled grasps. In terms of representation approaches, 3D Diffuser Actor [34] conditions on tokenized 3D scene representations rather than pooled embeddings, while R&D methods [71] implement a unified image-action framework with Vision Transformer architecture. For sequential task capabilities, ALOHA Unleashed [70] utilizes a transformer-based learning architecture with diffusion loss for bi-manual manipulation, whereas ChainedDiffuser [60] employs a two-stage approach that generates end-effector keyposes followed by connecting trajectories. Extending this concept, HDP [41] adds kinematics-aware capabilities for more physically realistic manipulation. Regarding category-level generalization, S²-Diffusion [52] integrates visual foundation models for spatial semantic understanding, while C3DM [138] introduces constrained-context conditioning with a novel fixation step for robustness against distractors. For long-horizon tasks, Diffuser [40] presents a unified sampling-planning approach with probabilistic behavior synthesis, and complementarily, DTP [32] provides 2D trajectory guidance in a two-phase framework. Furthermore, StructDiffusion [114] combines object-centric transformers with diffusion models for semantic rearrangement based on language goals, while PlayFusion [36] implements discrete bottlenecks for language-annotated skill acquisition. Additionally, ROSIE [65] augments robot learning by leveraging text-to-image diffusion models for data augmentation, and Scaling Up and Distilling Down [130] extends language conditioning capabilities across multiple tasks. Finally, GSC [95] proposes a probabilistic framework to chain learned skills with classifier-based guidance for constraints, and DBC [139] enhances policy optimization through diffusion-based behavioral cloning by modeling expert state-action pairs.

Generate Subgoal. The second category of methods generates meaningful intermediate subgoals that bridge high-level planning with low-level execution in robotic manipula-

tion. CoTDiffusion [75] introduces a hierarchical framework translating multi-modal prompts into visual subgoal images for chain-of-thought reasoning. ReflectVLM [42] implements reflective planning through diffusion-based dynamics models for iterative action refinement based on visual predictions. For longer-horizon tasks, Subgoal Diffuser [72] presents a coarse-to-fine approach dynamically determining subgoal density based on learned reachability estimates, while SuSIE [74] leverages pre-trained image-editing diffusion models to generate hypothetical future observations with superior generalization compared to language-conditioned policies. In scene arrangement tasks, DALL-E-Bot [140] employs text-to-image diffusion for generating target configurations representing natural object arrangements. CACTI [73] introduces a scalable framework incorporating generative models for multi-task data augmentation, while UniPi [37] leverages text-guided video generation to synthesize trajectories translated into actions through inverse dynamics models. Collectively, these approaches demonstrate how diffusion models effectively decompose complex tasks into interpretable, visually-guided subgoals, enhancing both performance and generalization capabilities.

Generate Robot Data. Diffusion models have emerged as powerful tools for addressing data scarcity in robotics through synthetic data generation. AdaptDiffuser [76] introduces a self-evolving framework generating high-quality synthetic expert data for goal-conditioned tasks through reward-guided diffusion. DemoGen [77] presents a fully synthetic demonstration generation approach adapting action trajectories to novel object configurations from limited human demonstrations, leveraging 3D point cloud manipulation for spatial augmentation. JUICER [78] offers a pipeline for learning high-precision manipulation by combining diffusion policies architectures with mechanisms for automatically expanding data coverage around bottleneck states. For simulation-to-reality transfer, ALDM-Grasping [79] employs adversarially supervised layout-to-image diffusion models to enhance simulation environments with photorealism, minimizing the gap between simulated and real-world grasping actions. ROSIE [65] leverages text-to-image diffusion models for aggressive data augmentation, using image inpainting guided by language prompts to introduce unseen objects and environments. Similarly, GenAug [80] employs pre-trained text-to-image generative models to create semantically meaningful visual augmentations while maintaining physical consistency with the original scene. In summary, these approaches demonstrate how diffusion models can substantially expand training data diversity and quantity while reducing extensive real-world data collection needs and improving generalization to novel scenarios.

Utilizing MoE Architecture. MoE [136] architectures combined with diffusion models offer powerful frameworks for enhancing robotic manipulation through specialized component integration. Sparse Diffusion Policy [81] introduces a sparse, reusable framework incorporating MoE into transformer-based diffusion policies, enabling selective activation of task-specific experts during execution. This architecture promotes computational efficiency by activating only relevant skills, facilitates knowledge transfer through skill reuse, and allows flexible incorporation of new experts

for continuous adaptation. This approach demonstrates how MoE architectures can effectively decompose complex robot control problems into specialized components while maintaining overall coherence, enhancing both performance and computational efficiency across diverse embodiments and robotic manipulation tasks.

Discussion. Integrating small-scale CNN or Transformer models with diffusion models has enhanced robotics manipulation in several ways. Direct action generation methods excel in real-time execution but may struggle with complex planning. Subgoal generation approaches provide hierarchical structure for long-horizon tasks, though determining optimal subgoal density remains challenging. Robot data generation methods address scarcity by synthesizing diverse datasets, potentially bridging sim-to-real gaps, despite limitations in capturing real-world variability. MoE architectures improve efficiency by activating only task-relevant model components, though effective expert routing remains difficult. In summary, these approaches address flexibility, efficiency, and generalization challenges, with future work needed on scalability, robustness, and adaptability for uncertain environments.

4.3 VAE / VQ-VAE Based Diffusion Policies

VAE [23] and VQ-VAE [24] approaches combined with diffusion models establish effective frameworks for robotic manipulation through latent space compression and structured action generation. GEVRM [82] introduces goal-guided diffusion policies using 2D and 3D VAEs to compress robot image states, with a T5 model [141] encoding language instructions and DiT modules generating goal states through prototypical contrastive learning. Discrete Policy [83] addresses multi-task manipulation by employing vector quantization to map action sequences into discrete latent spaces, creating task-specific codes that are reconstructed into actions conditioned on observations and language, while using diffusion for denoising in discrete rather than high-dimensional feature spaces. VPDD [33] bridges human and robot domains by compressing videos into unified tokens via VQ-VAE and employing discrete diffusion with mask-and-replace strategies to predict future video tokens, enabling fine-tuning on limited robot data guided by imagined future videos. Similarly, LAPA [53] learns from internet-scale videos without robot action labels by training an action quantization model with VQ-VAE objectives to extract discrete latent actions between frames, then pretraining a 7B-parameter latent VLA model [111] before fine-tuning on limited robot data to map from latent to robot actions.

Discussion. VAE/VQ-VAE architectures with diffusion models enhance robotic manipulation through dimensionality reduction and discrete representation learning. These approaches create structured latent spaces capturing fundamental action primitives, enabling efficient learning and knowledge transfer from human demonstrations. Discrete representations improve interpretability and support systematic exploration across tasks. Challenges include balancing reconstruction quality with latent space compactness, managing computational demands of concurrent VAE and diffusion training, and mitigating quantization errors.

5 DIFFUSION STRATEGY

Diffusion policies have demonstrated remarkable capabilities in robot manipulation, yet various strategies can further enhance their performance and applicability. This section examines five key directions for advancing diffusion models in robotics: 1) incorporating reinforcement learning to optimize policies through reward signals, 2) integrating equivariance principles to handle geometric transformations, 3) developing accelerated sampling or denoising strategies to improve computational efficiency, 4) designing improved classifier guidance techniques to enhance output generation while satisfying specific constraints, and 5) leveraging self-supervised learning approaches to enhance representation quality with unlabeled data. Each direction addresses specific challenges in robotic manipulation while offering complementary benefits to the diffusion framework.

Incorporating Reinforcement Learning. Reinforcement learning problems are formally described as Markov Decision Processes (MDPs) [142], consisting of essential elements: the state space S , the action space $A \subseteq \mathbb{R}^p$, the state transition probability function $P(s'|s, a)$, reward function $R(s, a, s')$, discount factor $\gamma \in (0, 1]$, and initial state distribution $d_0(\cdot)$.

The primary objective of reinforcement learning is to find an optimal policy π^* that selects actions maximizing the expected cumulative discounted reward:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]. \quad (10)$$

However, traditional reinforcement learning methods often struggle to represent complex policy distributions effectively. Conventional approaches, such as PPO [143] and TRPO [144], rely on unimodal distributions (e.g., Gaussian distributions), limiting their capability to model multimodal action patterns. This limitation frequently leads to convergence to local optima, restricted exploration, and compromised generalization in complex environments.

To overcome these limitations, recent studies have introduced diffusion models into reinforcement learning. For instance, Diffuser [40] formulates trajectory optimization as a denoising diffusion process, progressively refining noisy trajectories into optimized solutions. This approach significantly improves performance on long-horizon and adaptive tasks by enabling richer policy representation.

While single-task RL aims to determine an optimal policy π^* for maximizing expected long-term returns, multi-task reinforcement learning (Multi-task RL) generalizes this by defining a task distribution $T \sim p(T)$. Each task T corresponds to a specific MDP:

$$\mathcal{M}^{(T)} = (S^{(T)}, A, P^{(T)}, R^{(T)}, \mu^{(T)}, \gamma). \quad (11)$$

Rather than solving each task independently, multi-task RL seeks a universal policy π^* maximizing expected cumulative rewards across all tasks:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{T \sim p(T)} \left[\mathbb{E}_{a_t \sim \pi} \left(\sum_{t=0}^{\infty} \gamma^t r_t^{(T)} \right) \right], \quad (12)$$

where $r_t^{(T)}$ is the reward for task T at timestep t .

Integrating diffusion models into multi-task RL further enhances offline decision-making in complex scenarios.

MTDIFF [84] addresses multi-task learning as a trajectory denoising problem, effectively improving generalization and mitigating distribution shifts commonly encountered in Temporal Difference (TD) methods [145]. Moreover, it generates high-quality synthetic data to bolster the training process. Meanwhile, EDGI [86] unifies model-based reinforcement learning and planning as conditional generative modeling. Utilizing equivariant diffusion models, EDGI achieves efficient planning and superior generalization in object manipulation and navigation tasks. Similarly, Diffusion Reward [85] optimizes generative diversity to facilitate expert-like behaviors. It leverages pre-trained video diffusion models to construct dense reward functions, combining novelty-driven exploration with sparse rewards to enhance learning efficiency.

Diffusion models have notably improved complex policy parameterization and optimization in RL. For example, DIPO [146], the first online model-free diffusion-based RL method, uses stochastic diffusion processes to represent complex multimodal action distributions, overcoming traditional RL's limitations with unimodal policies. Building upon DIPO's foundation, DPPO [39] integrates diffusion-specific denoising steps into policy gradient optimization. This structured approach significantly enhances training stability and efficiency, proving particularly effective in fine-tuning policies for complex robotic manipulation tasks. While DIPO prioritizes expressive policy representation, DPPO further refines optimization robustness, facilitating reliable policy learning in high-dimensional action spaces.

Furthermore, latent diffusion models represent trajectory sequences through compressed latent skills [147], [148], facilitating Q-function learning, mitigating extrapolation errors, and effectively managing multimodal data distributions. Q-Score Matching [149] tightly integrates diffusion-based policies with Q-functions by iteratively aligning policy score functions and Q-function gradients, substantially improving optimization efficiency. Additionally, RLDiff-ORL [147] explores conditional generative models' rich latent spaces, achieving notable performance within batch-constrained Q-learning frameworks. These advances collectively highlight the significant potential of latent diffusion models for addressing the challenges inherent in complex reinforcement learning scenarios.

Combined with Equivariance. With the powerful generative capabilities of diffusion models, significant achievements have been made in fields such as image and text generation. Extending these models to SE(3) space, enabling them to handle rigid body transformations while preserving equivariance, has become a research hotspot in robotics and computer vision. Diffusion-EDFs [87], the first work on SE(3)-equivariant diffusion models for visual robotic manipulation, proposes a novel bi-equivariant diffusion model that utilizes a learned bi-equivariant score function for denoising, which is rigorously constrained to maintain SE(3) equivariance. Mathematically, the score function is defined as:

$$\rho(x|O_e) = \sum_{q \in Q(O_e)} w(x|O_e) \delta^{(3)}(x - q), \quad (13)$$

where $Q(\cdot) : O_e \rightarrow \{q_n\}_{n=1}^{N_q}$ is the query point function, which outputs the set of N_q query points, and

$w(\cdot|O_e) : \mathbb{R}^3 \rightarrow \mathbb{R}_{\geq 0}$ is the query weight field that assigns weights to each query point. In contrast, EDGI [86] proposes equivariant blocks and integrates them with a U-Net architecture, significantly improving performance in low-data regimes—matching the performance of the best non-equivariant baseline while requiring an order of magnitude less training data. The EquiBot [57] improves CNN-based diffusion policies by incorporating SIM(3)-equivariance. It extracts features from input by a point cloud encoder based on PointNet [150] and applies an SO(3)-equivariant U-Net for denoising. The model then restores scale and position to ensure equivariant action generation, enabling the robot to learn policies more efficiently and robustly while adapting to varying object poses and scales.

Accelerated Sampling or Denoising Strategies. Diffusion models, while powerful for generating complex distributions, face challenges in real-time robotic applications due to their inherently iterative sampling process. Numerous approaches tackle this computational bottleneck through accelerated denoising techniques. BESO [88] decouples score model learning from inference, enabling goal-conditioned behavior generation with merely three denoising steps, compared to the typical requirement of more than 30. Similarly, ManiCM [59] imposes consistency constraints on the diffusion process for one-step inference through consistency distillation, directly predicting action samples rather than noise for faster convergence in low-dimensional action manifolds. ImitDiff [90] implements this concept by incorporating a consistency policy into the DiT [19] action head, achieving high-frequency system responses via single-step denoising. Hierarchical approaches offer another solution, with DiffuserLite [89], ChainedDiffuser [60], and RK-Diffuser [41] introducing coarse-to-fine generation processes that reduce redundant information modeling and increase decision frequency. ReorientDiff [30] demonstrates effective performance using only 20 sampling steps through FastDPM [151] strategies. For practical applications, RoboKeyGen [91] decomposes the high-dimensional robot pose estimation problem into manageable 2D and 3D keypoint tasks while utilizing a fast DDIM sampler. Research on human behavior imitation [152] introduces alternative sampling schemes (Diffusion-X and Diffusion-KDE) while notably revealing that Classifier-Free Guidance impedes action generation despite its benefits in image generation. When integrating with reinforcement learning, DPPO [39] optimizes diffusion noise schedules for stability and exploration, while Diff-Dagger [92] improves convergence by employing identical diffusion steps during both training and inference. Alternative approaches explore flow matching [122], [123], [124], [125], [126], with π_0 [64] utilizing optimal transport displacement interpolation to define conditional probability paths that offer faster training, sampling, and improved generalization compared to traditional diffusion methods. These diverse approaches collectively address the computational challenges of diffusion models in robotic applications, making real-time performance increasingly attainable.

Employing Classifier (free) Guidance. Classifier guidance techniques [153] significantly enhance diffusion models' ability to generate outputs that satisfy specific constraints or conditions in robotics applications. In standard diffusion models, the denoising process iteratively trans-

forms noise into samples using a learned noise prediction function $\epsilon_\theta(x_t, t)$. Classifier guidance augments this process by incorporating a gradient direction from a classifier $p(y|x)$ to steer sampling toward regions with high probability of the desired condition:

$$\hat{\epsilon}_\theta(x_t, t, y) = \epsilon_\theta(x_t, t) - \sqrt{1 - \alpha_t} \nabla_{x_t} \log p(y|x_t), \quad (14)$$

where α_t controls the noise schedule and y represents the desired constraint or condition. Classifier-free guidance [106] eliminates the need for an explicit classifier by training a conditional model $\epsilon_\theta(x_t, t, y)$ alongside an unconditional model $\epsilon_\theta(x_t, t)$, combining them during sampling:

$$\hat{\epsilon}_\theta(x_t, t, y) = \epsilon_\theta(x_t, t) + w[\epsilon_\theta(x_t, t, y) - \epsilon_\theta(x_t, t)], \quad (15)$$

where $w \geq 0$ is a guidance scale controlling the strength of conditioning. DGDM [94] extends classifier guidance to balance diversity and task-specific requirements in sensorless manipulator design, while SkillDiffuser [93] combines skill abstractions with conditional diffusion planning, using classifier-free diffusion guidance to align trajectory sampling with desired features. BESO [88] implements goal-classifier-free guided policy by combining score-based diffusion policies with classifier-free guidance for sequential decision-making. Notably, in contrast to image generation, classifier-free guidance in sequential environments may encourage less common actions paired with specific observations rather than enhancing adherence to prompts. ReorientDiff [30] adopts a hybrid approach integrating generic classifier-free conditional sampling with classifier-guided sampling for object manipulation, mapping scene and language information into joint scene-task representations conditioning the diffusion model. GSC [95] offers a probabilistic framework that composes learned distributions during inference to generate long-horizon plans, with its diffusion model-based formulation facilitating constraint incorporation through both implicit in-painting and explicit classifier-based guidance, creating a flexible sampling strategy capable of handling multiple planning constraints simultaneously while maintaining probabilistic consistency.

Integration with Self-Supervised Learning. Self-supervised learning mechanisms enhance diffusion policies by improving representation learning without requiring additional labeled data. MDT [68] leverages a diffusion-based multimodal transformer backbone with self-supervised auxiliary objectives to master long-horizon manipulation tasks based on multimodal goals with minimal language annotations. By integrating pre-trained foundation models and implementing latent token alignment, MDT effectively processes diverse input modalities, allowing the model to handle complex tasks even with sparse language guidance. Similarly, Crossway Diffusion [96] introduces an effective approach that integrates auxiliary self-supervised learning (SSL) into diffusion-based visuomotor policy learning. Its carefully designed state decoder reconstructs original image pixels and state information from intermediate representations produced during the reverse diffusion process. By jointly optimizing both the self-supervised learning objective and the original diffusion loss, Crossway Diffusion develops more robust visual representations that capture meaningful environmental features.

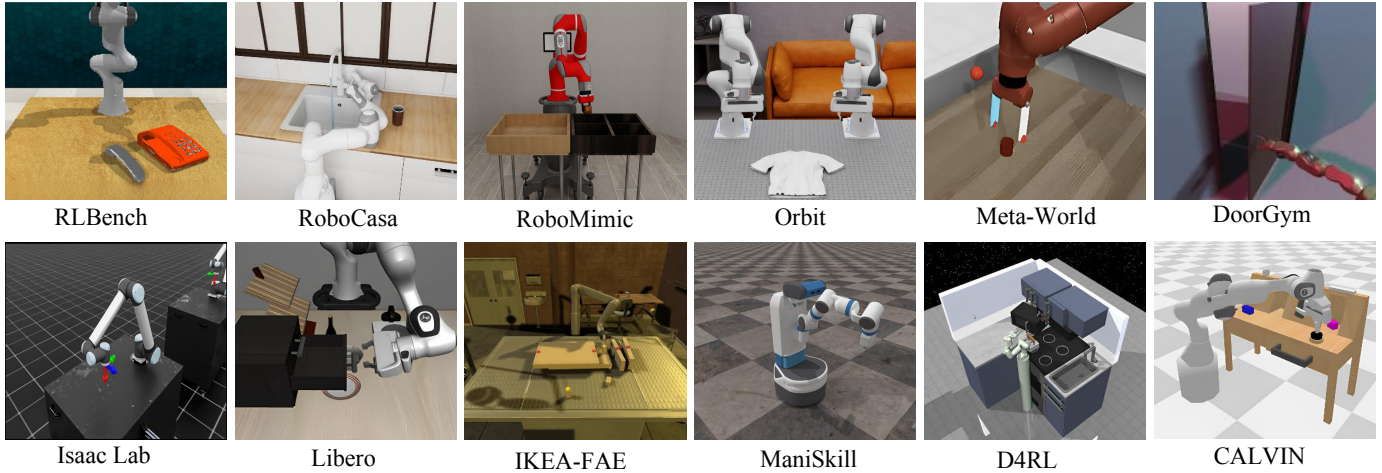


Fig. 6: Simulation robot platforms used in diffusion policies for robotic manipulation.

Discussion. This section explores five strategic directions advancing diffusion policies for robot manipulation. Incorporating reinforcement learning enables effective representation of complex policy distributions, improving performance in long-horizon and multi-task environments. Equivariance integration allows diffusion models to handle geometric transformations while preserving invariance properties, enhancing sample efficiency and generalization. Accelerated sampling techniques address computational bottlenecks through consistency distillation, one-step inference, and hierarchical methods, making diffusion models viable for real-time applications. Classifier guidance mechanisms enhance output generation to satisfy specific constraints, balancing diversity and task requirements. Self-supervised learning approaches improve representation quality without additional labeled data, yielding robust visual representations for complex tasks. Future research may focus on combining these approaches to develop unified frameworks addressing efficiency, representation power, and generalization across diverse robotic tasks.

6 ROBOT MANIPULATION BENCHMARK

To facilitate rigorous evaluation and fair comparison of diffusion policies for robotic manipulation, we have systematically collected and organized a comprehensive set of benchmarks encompassing both simulation environments and real-world robots. This collection represents the current landscape of testing infrastructures used across the field, providing researchers with valuable reference points for experimental design and performance assessment.

6.1 Simulation

Simulation environments play a crucial role in robotic manipulation, serving as essential tools for developing, validating, and evaluating algorithms. Figure 6 provides a comprehensive overview of diverse robotic simulation environments. Different platforms exhibit distinct characteristics in terms of physical simulation accuracy, computational efficiency, task complexity, and interaction capabilities, making them suitable for diverse research needs. Table 2 provides

a comprehensive comparison of these platforms, detailing their key specifications and dataset configurations to facilitate a systematic evaluation of their suitability for different robotic manipulation tasks.

Physics engines form a critical distinguishing element among simulation platforms, determining both simulation accuracy and computational efficiency. Several prominent engines include MuJoCo [171], which provides high-precision physics modeling and serves as the foundation for benchmarks like Meta-World [154] and D4RL [166]; PhysX [172], a GPU-accelerated engine implemented in Isaac Gym [173] and ManiSkill3 [174], which enables highly parallelized simulations; PyBullet [175], an open-source solution offering real-time physics capabilities integrated into environments such as CALVIN [169] and OpenAI Gym [176]; and CoppeliaSim (formerly V-REP) [177], which excels in modeling complex robotic interactions, as demonstrated in RL Bench [155]. Some platforms incorporate multiple physics engines to optimize the trade-off between accuracy and performance. Isaac Sim [178], for instance, utilizes PhysX [172] to deliver high-fidelity physics simulations with GPU acceleration, making it particularly suitable for industrial robotics applications and large-scale AI training. Similarly, Gazebo [179] offers flexibility through its support for multiple engines including ODE [180], Bullet [175], and PhysX [172], thereby facilitating research in robotic navigation and SLAM within the ROS ecosystem.

Beyond physics simulation, platforms vary significantly in task complexity and interaction capabilities. Benchmarks such as Meta-World [154] and D4RL [166] focus on standardized robotic manipulation tasks, providing controlled environments for reinforcement learning evaluation. In contrast, RL Bench [155] offers over 100 multi-phase manipulation tasks, demanding higher levels of perception and control. CALVIN [169] introduces language- and vision-driven task descriptions, advancing research in multimodal learning for robotic control.

Another critical factor is computational efficiency, as some platforms are designed for scalable reinforcement learning training. Isaac Gym [173] and ManiSkill3 [174] exploit GPU parallelization via PhysX [172], allowing efficient large-scale simulations that accelerate policy train-

TABLE 2: Simulation Robot Platform Specifications. This table provides an overview of various simulation platforms utilized in diffusion policies for robotic manipulation, highlighting their physics engines, robot models, perspectives, task diversity, and total episodes used in training datasets.

Platform	Physics Engine	Robot	Views	Tasks Episodes	
Meta-World [154]	MuJoCo	4-DOF Sawyer	behindGripper, corner, topview, gripperPOV	50	5K
RLBench [155]	V-REP	7-DOF UR5/6-DOF Mico 6-DOF Jaco/6-DOF Sawyer	left_shoulder, right_shoulder, wrist, front	18	1.8K
DEDO [156]	PyBullet	7-DoF Franka	sideview, eye_in_hand	2	–
Libero [157]	MuJoCo	7-DOF Franka	frontview, birdview, agentview, sideview	130	6.5K
DoorGym [158]	PyBullet	8-DoF BLUE, 7-DoF BLUE 14-DOF Baxter	static, gripperPOV left shoulder, right shoulder, wrist, front	3	9k
RoboMimic [159]	MuJoCo	7-DOF Franka	agentview, eye_in_hand	8	1.6K
Isaac Lab [160]	PhysX, MuJoCo, Warp	7-DOF Franka 6-DOF UR5/10/16	agentview, topview, eye_in_hand, sideview	5	–
RoboCasa [161]	MuJoCo	12-DOF Franka	center, left, right, frontview, eye_in_hand	100	5K
ManiSkill3 [162]	PhysX	7-DOF Franka/6-DOF WidowX-250S Mobile Base & 7-DoF Fetch	agentview, eye_in_hand	20	30K
RoboCAS [163]	PhysX	7-DOF Franka	gripperPOV, agentview, static	3	7.3K
IKEA-FAE [164]	ODE	8-DoF Franka, 7-DoF Fetch 14-DOF Baxter 6-DOF Jaco	–	80	–
Colosseum [165]	PyRep	7-DOF Franka	left_shoulder, right_shoulder, wrist, front	20	2K
D4RL [166]	MuJoCo	7-DOF Franka	–	4	0.8K
Gymnasium [167]	MuJoCo	9-DoF Franka Mobile Base & 7-DoF Fetch	agentview, gripperPOV, eye_in_hand	7/4	–
OmniIsaacGymEnvs [168]	PhysX	7-DOF Franka	agentview, topview, eye_in_hand, sideview	5	–
CALVIN [169]	PyBullet	7-DOF Franka	static, gripperPOV	34	20K
Orbit [160]	PhysX	7-DOF LBR iiwa, 6-DOF UR10 Mobile Base & 7-DoF Franka	agentview, topview, eye_in_hand, sideview	–	–
Habitat Lab [170]	PyBullet	7-DoF Franka Mobile Base & 7-DoF Fetch/Stretch	eye_in_hand, agentview, sideview agentview, sideview	6	70k

ing. SAPIEN [186], which combines Bullet [175] and MuJoCo [171], not only achieves high-precision modeling but also provides support for large-scale robotic interaction tasks. Conversely, platforms such as Gazebo [179] and Webots [187] prioritize accurate robotic control and kinematic modeling, often at the cost of lower computational speed, making them better suited for robot navigation, SLAM, and motion planning rather than for reinforcement learning.

In summary, simulation platforms cater to different research needs depending on their physics simulation accuracy, task complexity, computational efficiency, and application domains. While policy learning environments such as Meta-World [154], D4RL [166], and ManiSkill3 [174] emphasize modularity and scalability, robotic manipulation and interaction environments like RLBench [155] and RoboMimic [159] focus on fine-grained task execution. Meanwhile, industrial and mobile robot simulation platforms such as Isaac Sim [178], Gazebo [179], and Webots [187] provide high-fidelity environments for real-world deployment validation. Understanding these differences allows researchers to select the most suitable platform based on specific experimental objectives.

6.2 Real World

We provide a comprehensive overview of diverse real-world robots in Figure 7, and delineate their critical specifications including degrees of freedom, workspace range, payload capacity, end-effectors, and integrated sensors in Table 3. As advanced embodied intelligence platforms, these real-world robots transcend mere mechanical constructs, emerging as sophisticated agents capable of perceiving, reasoning, and dynamically interacting with complex environments. They serve as critical research infrastructures for investigating pivotal domains of intelligent systems, encompassing multi-modal perception, reinforcement learning control, end-to-end decision-making, and innovative cloud-edge collaborative computing paradigms.

These capabilities are vividly embodied in cutting-edge robotic platforms that bridge theoretical research with real-world applications. The Fourier GR1 [71], exemplifying this technological sophistication, is strategically equipped with a ring-shaped microphone and a Realsense depth camera, enabling nuanced multi-modal perception and comprehensive data collection. Concurrently, the ALOHA [182] system

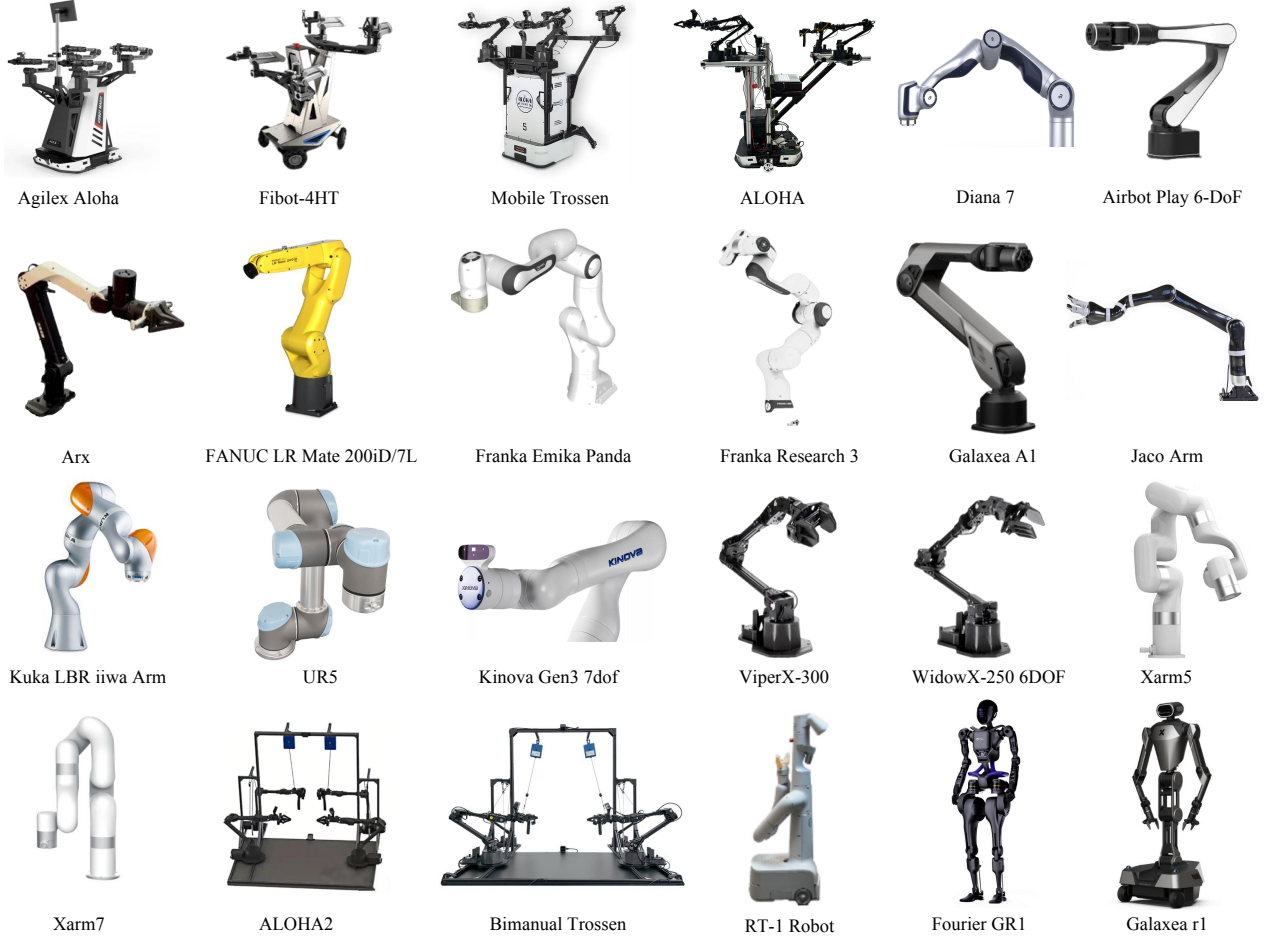


Fig. 7: Real-world robots used in diffusion policies for robotic manipulation.

integrating robotic arms with mobile bases represents a groundbreaking, low-cost approach to end-to-end imitation learning, facilitating direct knowledge transfer from real-world demonstrations. This platform provides an invaluable framework for examining perception-action feedback loops in embodied intelligence research.

Notably, robots featuring interchangeable end effectors, such as the UR5 [181], Xarm7 [61], and Kinova Gen3 [57], epitomize computational flexibility and adaptability. These systems enable researchers to explore sophisticated robot learning methodologies and dexterous manipulation techniques, allowing dynamic reconfiguration to address diverse operational scenarios and task requirements. As the field of embodied artificial intelligence continues its rapid evolution, these robotic platforms will serve as cornerstone technologies, driving the development of increasingly intelligent, adaptive, and context-aware systems poised to revolutionize real-world applications.

7 SUGGESTIONS & FUTURE DIRECTIONS

The rapid development of diffusion policies for robotic manipulation requires a reasonable way to method selection and identification of promising research directions. In this section, we first provide practical recommendations for selecting appropriate diffusion policy approaches based on

specific requirements and constraints, followed by an exploration of future research directions that could significantly advance the field.

7.1 Recommendations on diffusion policies selection

Based on our systematic analysis of diffusion policy methods for robotic manipulation, we present the following recommendations. These guidelines assist researchers and practitioners in selecting appropriate approaches based on their specific requirements and constraints.

Task Complexity and Data Availability Trade-offs. For complex manipulation tasks with abundant robot data, direct action generation methods such as PlayFusion [36] and DiT-Block Policy [107] provide superior performance by accurately capturing multimodal action distributions. When robot data is limited, methods incorporating equivariance principles (Diffusion-EDFs [87], EquiBot [57]) or leveraging pre-trained foundation models (Octo [27], RDT-1B [28]) enhance sample efficiency. For extremely data-scarce scenarios, human video-based methods such as LAPA [53], VPDD [33], or Human2Robot [55] utilize extensive internet demonstrations before fine-tuning on limited robot data, substantially improving model generalization across diverse manipulation tasks.

Computational Efficiency Considerations. When real-time performance is essential, methods employing accelerated sampling strategies such as ManiCM [59], ImitDiff [90],

TABLE 3: **Real-World Robot Specifications.** This table provides an overview of various real-world robots in diffusion policies for robotic manipulation, highlighting their names, types, degrees of freedom (DoF), workspace ranges, maximum payload capacities, end effectors, and sensor configurations.

Robot Name	Robot Type	DoF	Workspace Range (mm)	Max Payload (kg)	End Effector	Sensors
WidowX-250 6DOF [74]	Single Arm	6	650	0.25	Gripper	–
galaxea r1 [77]	Upper Body & Mobile Base	14	1400	5 (Single Arm)	Gripper	binocular stereo camera
UR5 [181]	Single Arm	6	850	5	Interchangeable	–
RT-1 Robot [65]	Single Arm & Mobile Base	7	–	–	Gripper	camera
ViperX-300 [27]	Single Arm	6	750	0.75	Gripper	–
Xarm7 [61]	Single Arm	7	700	3.5	Interchangeable	camera
Xarm5 [80]	Single Arm	5	700	3	Interchangeable	camera
Franka Emika Panda [41]	Single Arm	7	850	3	Interchangeable	–
Franka Research 3 [54]	Single Arm	7	855	3	Interchangeable	–
Airbot Play 6-DoF [90]	Single Arm	6	647	3.5	Gripper	–
Kinova Gen3 7-DoF [57]	Single Arm	7	902	4	Interchangeable	camera
FANUC LR Mate 200iD/7L [81]	Single Arm	6	911	7	Interchangeable	–
ALOHA [182]	Dual Master & Slave Arms & Mobile Base	6	750	0.75 (Single Arm)	Gripper	camera
ALOHA2 [183]	Dual Master & Slave Arms	6	750	0.75 (Single Arm)	Gripper	camera
Jaco Arm [114]	Single Arm	7	–	–	Three-Finger Gripper	–
Galaxea A1 [184]	Single Arm	6	700	5	Interchangeable	–
Kuka LBR iiwa Arm [72]	Single Arm	7	800	7	Interchangeable	–
Mobile Trossen [64]	Dual Master & Slave Arms & Mobile Base	6	750	0.75 (Single Arm)	Gripper	camera
Fibot-4HT [64]	Dual Master & Slave Arms & Mobile Base	6	744	3 (Single Arm)	Gripper	camera
Diana 7 [185]	Single Arm	7	923	7	Interchangeable	–
Bimanual Trossen [64]	Dual Master & Slave Arms	6	750	0.75 (Single Arm)	Gripper	camera
Arx X5 [64]	Single Arm	6	620	1.5	Gripper	–
AgileX Aloha [28]	Dual Master & Slave Arms & Mobile Base	6	–	0.75 (Single Arm)	Gripper	camera
Fourier GR1 [56]	Humanoid	–	1680	3 (Single Arm)	Dexterous Hand	Ring-shaped Microphone realsense camera

or DiffuserLite [89] achieve one-step or few-step inference, significantly increasing model inference speed and action generation frequency. For resource-constrained deployment environments, compact CNN or Transformer architectures like ALOHA Unleashed [70] or diffusion policies [26] offer an optimal balance between performance and computational efficiency, effectively meeting real-time action generation requirements.

Spatial Understanding Requirements: When precise spatial reasoning is critical, 3D representation methods (DP3 [31], 3D Diffuser Actor [34]) outperform 2D approaches, particularly for manipulation environments with complex object arrangements or occlusions. For applications requiring invariance to object position and orientation, equivariant models (Diffusion-EDFs [87], EquiBot [57]) demonstrate robust performance across diverse spatial scenarios, enabling cross-task generalization through limited demonstrations. Multi-view point cloud methods (DN-Act [61], Bi3D Diffuser Actor [29]) are particularly effective for manipulation tasks with high spatial awareness requirements, providing comprehensive environmental understanding, despite increased model complexity and computational demands.

Multi-task and Heterogeneity Scenarios. In multi-task learning applications, VAE/VQ-VAE integration methods (Discrete Policy [83], GEVRM [82]) enable efficient task-specific latent encoding, facilitating knowledge transfer between related tasks. MoE-based approaches (SDP [81]) excel in complex multi-task scenarios through dynamic expert selection mechanisms, enabling efficient skill reuse and sharing across different tasks. When addressing heterogeneous robot environments, models specifically designed for embodiment perceptual heterogeneity (π_0 [64], RDT-1B [28]) provide essential cross-entity adaptation capabilities.

Language-guided Manipulation. When complex language instruction understanding is required, large language models equipped with diffusion heads (Octo [27], Diffusion-VLA [69], ChatVLA [134]) effectively bridge semantic understanding and precise action generation. For applications with high instruction following efficiency requirements, lightweight multimodal models with language following capabilities (MDT [68], S²-Diffusion [52]) offer a practical balance of performance and resource utilization. Compositional or sequential reasoning tasks benefit from combining LLMs for planning with diffusion models for control (HiP [66], ROSIE [65]).

Planning Horizon and Task Structure. For long-horizon tasks with clear hierarchical structures, subgoal generation approaches (Subgoal Diffuser [72], Plan Diffuser [67], ReflectVLM [42]) effectively decompose complex problems while maintaining end-to-end trainability. When addressing exploration challenges in reinforcement learning, diffusion-based methods incorporating RL techniques (Diffuser [40], EDGI [86], Diffusion Reward [85]) demonstrate superior exploration capabilities and sample efficiency. For tasks requiring both exploration and precise execution, hybrid approaches combining classifier guidance with diffusion-based policies (GSC [95], ReorientDiff [30]) provide a balanced solution.

These recommendations offer valuable guidance for selecting diffusion policy methods based on key practical considerations. By evaluating task requirements and available resources, researchers and practitioners can identify optimal approaches for their specific robotic manipulation applications.

7.2 Future Directions

Based on our comprehensive research and analysis of diffusion policies for robotic manipulation, we identify several promising research directions that could significantly advance this field:

Integrating with LLMs. Develop more efficient integration techniques between diffusion-based action generation and LLMs, especially Multi-Modal Large Models (MMLMs) capable of processing various input sources simultaneously. The goal is to combine the powerful reasoning capabilities of LLMs with diffusion models' ability to model multi-modal distributions, fully exploiting LLMs' potential in robotic manipulation tasks and enhancing model generalization across different tasks and scenarios.

Embodied Large Models Based on Diffusion Policies. While existing LLMs are mostly designed for general tasks, developing specialized large models based on diffusion policies specifically for robotic manipulation represents a promising research direction. Such models could be pre-trained on existing robot data or human demonstrations and integrated with recent large model technologies.

Data Heterogeneity in Robotic Pre-training. Advance the development of universal action spaces to seamlessly transfer diffusion policies across heterogeneous data with diverse robot proprioceptions and sensor configurations. Future research should aim to develop unified pre-trained models that effectively accommodate and leverage diverse heterogeneous robotic datasets, learn general robotic manipulation skills, and enhance model generalization across different scenarios.

Leveraging Internet-Scale Human Video Data. Develop more sophisticated alignment techniques between human action spaces and robot action spaces to effectively transfer rich human videos to robotic manipulation, reducing dependence on robot data. For example, combine self-supervised learning methods to extract meaningful skills from unlabeled human videos without explicit action annotations, ultimately enhancing the model's action learning capabilities across diverse scenarios.

Theoretical Innovations in Diffusion Model. Investigate the intrinsic theoretical connections between diffusion

processes and optimal action generation methods to establish a more solid formal foundation for diffusion-based robotic manipulation. Improve diffusion model principles by developing specialized diffusion formulations that explicitly consider the inherent physical constraints and dynamic characteristics, creating new theoretical frameworks specifically designed for robotic manipulation tasks.

Enhancing Sampling and Inference Efficiency. Introduce the latest diffusion model acceleration or sampling techniques to compress multi-step diffusion processes into single-step or few-step models without sacrificing generation quality, significantly improving the real-time performance in generating robotic actions. Develop adaptive sampling strategies that allocate computational resources based on task complexity and required precision, achieving a balance between computational efficiency and accuracy in robotic manipulation tasks.

Integration with Reinforcement Learning. How to better combine diffusion models and reinforcement learning represents a promising direction. Develop unified frameworks that seamlessly integrate the generative capabilities of diffusion models with the optimality-seeking behavior of reinforcement learning. Research diffusion-based exploration strategies to generate accurate robotic manipulation actions in complex high-dimensional action spaces.

Real-World Deployment and Generalization. The sim-to-real gap remains a major challenge in robotic manipulation tasks. Develop simulation-to-real generalization methods based on diffusion policies, leveraging relatively accessible robot simulation data for training and transferring to real-world scenarios, achieving safer practical application deployment.

Overall, these future directions highlight the significant potential for advancement in diffusion policies for robotic manipulation. By addressing these challenges, researchers can progress toward more powerful, efficient, and generalizable robotic systems that fully leverage the unique advantages of diffusion models while overcoming their current limitations.

8 CONCLUSIONS

This survey provides the first comprehensive analysis of diffusion policies for robotic manipulation, organizing this rapidly evolving field across three key dimensions: robot data representation, model architecture, and diffusion strategy. Our analysis reveals how 2D representations offer computational efficiency while 3D representations provide enhanced spatial awareness, and how integrating heterogeneous data effectively addresses cross-embodiment challenges. We examine architectural approaches ranging from large language models with rich semantic understanding to computationally efficient CNN/transformer implementations and VAE/VQ-VAE frameworks for structured latent representations. Additionally, we analyze various diffusion strategies that incorporate reinforcement learning, equivariance principles, accelerated sampling techniques, classifier (free) guidance, and self-supervised learning to enhance performance across diverse manipulation tasks.

Furthermore, after thorough and comprehensive analysis, our taxonomy and recommendations offer researchers

and practitioners valuable guidance for selecting appropriate methods based on specific requirements and constraints. Looking ahead, we identify several promising research directions: integrating with LLMs, addressing heterogeneity in robotic data, theoretical innovations and inference acceleration in diffusion models, integration with reinforcement learning, leveraging internet-scale human demonstrations, embodied large models based on diffusion policies, and improving real-world deployment. By maintaining an updated repository of resources, we aim to support continued advancement in this field, ultimately progressing toward more capable, generalizable robotic manipulation systems that function effectively in complex real-world environments.

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